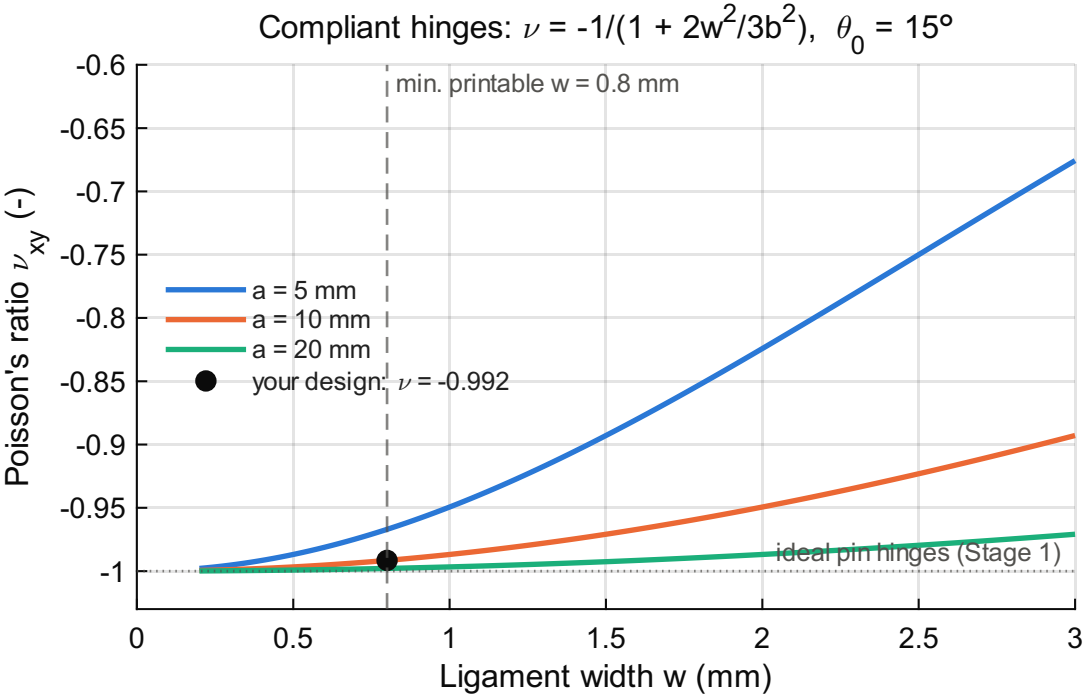
Compliant hinges: ν = -1/(1 + 2w^2/3b^2), θ_0 = 15°
min. printable w = 0.8 mm
a = 5 mm
a = 10 mm
a = 20 mm
your design: ν = -0.992
ideal pin hinges (Stage 1)
Poisson's ratio ν_{xy} (-)
Ligament width w (mm)
-0.6
-0.65
-0.7
-0.75
-0.8
-0.85
-0.9
-0.95
-1
0
0.5
1
1.5
2
2.5
3